

This searing desert is one of the main places on Earth where boron is found.



Death Valley, USA

Needle- and leaf-shaped crystals



Uses



This tough glass contains boron oxide.



These white crystals are obtained from sodium borate.

LCD screen



This screen is composed of boron-rich glass, which makes it scratch-resistant.

The protective body of this tank contains boron carbide, a compound of boron and carbon.

Military tank



THÉNARD AND GAY-LUSSAC

The salt sodium borate, also called borax, was already in use 1,000 years ago. In 1808, the Frenchmen Joseph Louis Gay-Lussac and Louis Jacques Thénard isolated pure boron by heating borax with potassium.



Louis Jacques Thénard

Born into a poor family, Thénard excelled as a scientist. He also discovered a compound called hydrogen peroxide.



Joseph Louis Gay-Lussac

This French chemist is also remembered for discovering that the pressure of a gas goes up with its temperature.

This clay is bouncy but also firm because it contains boron.

Modelling clay



Boron carbide is one of the **hardest** materials in use today.

Tough, heat-resistant glassware, such as **measuring cups**, are strengthened with boron. **Boric acid** is a natural antiseptic and can be used to treat minor cuts and scrapes. A flexible layer of boron-based glass fibres is used to toughen thin **LCD screens** for televisions and laptops. Even

some kinds of **modelling clay** and bouncy silly putty contain boron compounds. Boron is named after a crumbly white salt called borax, which is used in detergents. The element is also present in a diverse range of objects, from insecticides to armour for **military tanks**.